

PepperCone Physical Improvement Plan

Main idea

- The reflected image looks washed out and ghostly when ambient light and background clutter show through the cone film along with the intended image.
- Two physical changes to be made. Block ambient light and background around the rig (shadow box / shadow cage), and control how much unwanted light passes through the film (tinted cone).

Current setup

- Cone film reflects video from the display underneath it.
- No enclosure around the rig yet. Ceiling, desks, and people walking by are visible through the film in our current photos.

Research notes

Shadow box / shadow cage

- Ambient light and background detail compete directly with the reflected image. Keeping the surrounding area dark and reducing outside light increases the contrast of the illusion [1].
- Small DIY dark-box builds are a standard way to intensify the effect at tabletop scale, which matches the size of our rig.

Tinted cone / reflective film

- Commercial Pepper's Ghost film is built to be as transparent as possible, because higher transparency is what lets the image reflect while the film itself stays invisible to the audience [1][2].
- Film tension and flatness matters. Creases and slack scatter light and blur the reflection, so keeping the film wrinkle-free is treated as a core requirement, not tint.
- None of the sources we found recommend tinting the reflective film to fix contrast. Tinting reduces total light passing through in both directions, so it would dim the reflected image along with the background [1].

Plan

- Step 1: Build a simple shadow enclosure around the sides and back of the rig, leaving the front open for viewing.
- Step 2: Active the Pepper's Cone with the new enclosure and note the projection's quality
- Step 3: If contrast is still weak, test a lightly tinted film sample against the current untinted film, to see if the brightness tradeoff is worth it.
- Step 4: Separately, test a stiffer material to reduce visible creases.

Test notes

Compare:

- No enclosure
- Black fabric shadow box
- Black foam-core shadow box
- Tinted film sample (only if the shadow box alone is not enough)

Record:

- Photos/recording from the same angle and distance each time
- Ambient light condition
- Subjective rating of projection's quality
- Any visible wrinkles or creases in the film

Current decision

- Build the shadow box first. It is cheap, reversible, and matches what the research treats as standard practice.
- Hold off on tinting the film until we measure whether the shadow box alone solves the contrast problem, since tinting always costs some brightness.
- Look into a new material for the cone that could minimize creases and address a separate issue.

Links

1. [Gerriets Pepper's Ghost special effect film spec sheet](#)
2. [Reflective surface tensioning and cleaning system for pepper's ghost illusion](#)